

April 15, 2020

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Biosolids
Pace Project No.: 50253325

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on March 31, 2020. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,



Melanie Booms
melanie.booms@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Biosolids

Pace Project No.: 50253325

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268

Illinois Accreditation #: 200074

Indiana Drinking Water Laboratory #: C-49-06

Kansas/TNI Certification #: E-10177

Kentucky UST Agency Interest #: 80226

Kentucky WW Laboratory ID #: 98019

Michigan Drinking Water Laboratory #9050

Ohio VAP Certified Laboratory #: CL0065

Oklahoma Laboratory #: 9204

Texas Certification #: T104704355

West Virginia Certification #: 330

Wisconsin Laboratory #: 999788130

USDA Soil Permit #: P330-19-00257

REPORT OF LABORATORY ANALYSIS

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SAMPLE SUMMARY

Project: Biosolids

Pace Project No.: 50253325

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50253325001	Digester #3	Solid	03/30/20 13:45	03/31/20 09:45

REPORT OF LABORATORY ANALYSIS

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SAMPLE ANALYTE COUNT

Project: Biosolids
Pace Project No.: 50253325

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50253325001	Digester #3	EPA 8081	KAV	21	PASI-I
		EPA 8081	CPH	8	PASI-I
		EPA 8082	RID	8	PASI-I
		EPA 8151A	BJW	3	PASI-I
		EPA 6010	KJE	6	PASI-I
		EPA 6010	JPk	8	PASI-I
		EPA 6020	DMT	10	PASI-I
		EPA 7470	ILP	1	PASI-I
		EPA 7471	FRW	1	PASI-I
		EPA 8270	JCM	67	PASI-I
		EPA 8270	JCM	18	PASI-I
		EPA 8260	TMW	55	PASI-I
		EPA 5030/8260	ALA	13	PASI-I
		SM 2540G	SWJ	1	PASI-I
		SM 2540G	SWJ	1	PASI-I
		SM 2540G	MMS	1	PASI-I
		EPA 9038	TPD	1	PASI-I
		EPA 9045	SCM	1	PASI-I
		EPA 351.2	DAC1	1	PASI-I
		EPA 353.2	SLB	2	PASI-I
		SM 4500-NH3 G	DAC1	1	PASI-I
		SM 4500-Cl-E	GWA	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50253325

Sample: Digester #3 **Lab ID: 50253325001** Collected: 03/30/20 13:45 Received: 03/31/20 09:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8081 GCS Pesticide Solids								
Analytical Method: EPA 8081 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
Aldrin	<271	ug/kg	271	1	04/09/20 14:25	04/10/20 15:03	309-00-2	
alpha-BHC	<271	ug/kg	271	1	04/09/20 14:25	04/10/20 15:03	319-84-6	
beta-BHC	<271	ug/kg	271	1	04/09/20 14:25	04/10/20 15:03	319-85-7	
delta-BHC	<271	ug/kg	271	1	04/09/20 14:25	04/10/20 15:03	319-86-8	
gamma-BHC (Lindane)	<271	ug/kg	271	1	04/09/20 14:25	04/10/20 15:03	58-89-9	
Chlordane (Technical)	<5430	ug/kg	5430	1	04/09/20 14:25	04/10/20 15:03	57-74-9	
alpha-Chlordane	<271	ug/kg	271	1	04/09/20 14:25	04/10/20 15:03	5103-71-9	
gamma-Chlordane	<271	ug/kg	271	1	04/09/20 14:25	04/10/20 15:03	5103-74-2	
4,4'-DDD	<543	ug/kg	543	1	04/09/20 14:25	04/10/20 15:03	72-54-8	
4,4'-DDE	<543	ug/kg	543	1	04/09/20 14:25	04/10/20 15:03	72-55-9	
4,4'-DDT	<543	ug/kg	543	1	04/09/20 14:25	04/10/20 15:03	50-29-3	
Dieldrin	<543	ug/kg	543	1	04/09/20 14:25	04/10/20 15:03	60-57-1	
Endosulfan I	<271	ug/kg	271	1	04/09/20 14:25	04/10/20 15:03	959-98-8	
Endosulfan II	<543	ug/kg	543	1	04/09/20 14:25	04/10/20 15:03	33213-65-9	
Endosulfan sulfate	<543	ug/kg	543	1	04/09/20 14:25	04/10/20 15:03	1031-07-8	
Endrin	<543	ug/kg	543	1	04/09/20 14:25	04/10/20 15:03	72-20-8	
Endrin aldehyde	<543	ug/kg	543	1	04/09/20 14:25	04/10/20 15:03	7421-93-4	
Heptachlor	<271	ug/kg	271	1	04/09/20 14:25	04/10/20 15:03	76-44-8	
Heptachlor epoxide	<271	ug/kg	271	1	04/09/20 14:25	04/10/20 15:03	1024-57-3	
Toxaphene	<5430	ug/kg	5430	1	04/09/20 14:25	04/10/20 15:03	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	67	%.	10-129	1	04/09/20 14:25	04/10/20 15:03	2051-24-3	

8081 GCS Pest RV, TCLP

Analytical Method: EPA 8081 Preparation Method: EPA 3510

Leachate Method/Date: EPA 1311; 04/07/20 21:10 Initial pH: 8.35; Final pH: 5.16

Pace Analytical Services - Indianapolis

gamma-BHC (Lindane)	<0.00025	mg/L	0.00025	1	04/13/20 10:30	04/13/20 15:24	58-89-9	
Chlordane (Technical)	<0.0025	mg/L	0.0025	1	04/13/20 10:30	04/13/20 15:24	57-74-9	
Endrin	<0.00050	mg/L	0.00050	1	04/13/20 10:30	04/13/20 15:24	72-20-8	
Heptachlor	<0.00025	mg/L	0.00025	1	04/13/20 10:30	04/13/20 15:24	76-44-8	
Heptachlor epoxide	<0.00025	mg/L	0.00025	1	04/13/20 10:30	04/13/20 15:24	1024-57-3	
Methoxychlor	<0.0025	mg/L	0.0025	1	04/13/20 10:30	04/13/20 15:24	72-43-5	
Toxaphene	<0.0050	mg/L	0.0050	1	04/13/20 10:30	04/13/20 15:24	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	39	%.	10-113	1	04/13/20 10:30	04/13/20 15:24	2051-24-3	

8082 GCS PCB Solids

Analytical Method: EPA 8082 Preparation Method: EPA 3546

Pace Analytical Services - Indianapolis

PCB-1016 (Aroclor 1016)	<17600	ug/kg	17600	1	04/10/20 13:50	04/14/20 12:47	12674-11-2	
PCB-1221 (Aroclor 1221)	<17600	ug/kg	17600	1	04/10/20 13:50	04/14/20 12:47	11104-28-2	
PCB-1232 (Aroclor 1232)	<17600	ug/kg	17600	1	04/10/20 13:50	04/14/20 12:47	11141-16-5	
PCB-1242 (Aroclor 1242)	<17600	ug/kg	17600	1	04/10/20 13:50	04/14/20 12:47	53469-21-9	
PCB-1248 (Aroclor 1248)	<17600	ug/kg	17600	1	04/10/20 13:50	04/14/20 12:47	12672-29-6	
PCB-1254 (Aroclor 1254)	<17600	ug/kg	17600	1	04/10/20 13:50	04/14/20 12:47	11097-69-1	
PCB-1260 (Aroclor 1260)	<17600	ug/kg	17600	1	04/10/20 13:50	04/14/20 12:47	11096-82-5	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50253325

Sample: Digester #3 **Lab ID: 50253325001** Collected: 03/30/20 13:45 Received: 03/31/20 09:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
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8082 GCS PCB Solids

Analytical Method: EPA 8082 Preparation Method: EPA 3546
Pace Analytical Services - Indianapolis

Surrogates

Tetrachloro-m-xylene (S)	68	%	25-123	1	04/10/20 13:50	04/14/20 12:47	877-09-8	
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8151A CI Acid Herbicides TCLP

Analytical Method: EPA 8151A Preparation Method: EPA 8151
Leachate Method/Date: EPA 1311; 04/07/20 21:10 Initial pH: 8.35; Final pH: 5.16
Pace Analytical Services - Indianapolis

2,4-D	<0.0050	mg/L	0.0050	1	04/13/20 11:38	04/14/20 12:21	94-75-7	
2,4,5-TP (Silvex)	<0.0050	mg/L	0.0050	1	04/13/20 11:38	04/14/20 12:21	93-72-1	
Surrogates								
2,4-DCAA (S)	57	%	30-140	1	04/13/20 11:38	04/14/20 11:50	19719-28-9	

6010 MET ICP

Analytical Method: EPA 6010 Preparation Method: EPA 3050
Pace Analytical Services - Indianapolis

Aluminum	6160	mg/kg	1450	1	04/06/20 15:14	04/07/20 13:34	7429-90-5	
Calcium	22200	mg/kg	1450	1	04/06/20 15:14	04/07/20 13:34	7440-70-2	
Magnesium	3900	mg/kg	1450	1	04/06/20 15:14	04/07/20 13:34	7439-95-4	
Phosphorus	40400	mg/kg	1450	1	04/06/20 15:14	04/07/20 13:34	7723-14-0	N2
Potassium	2110	mg/kg	1450	1	04/06/20 15:14	04/07/20 13:34	7440-09-7	
Sodium	3330	mg/kg	1450	1	04/06/20 15:14	04/07/20 13:34	7440-23-5	

6010 MET ICP, TCLP

Analytical Method: EPA 6010 Preparation Method: EPA 3010
Leachate Method/Date: EPA 1311; 04/07/20 21:10 Initial pH: 8.35; Final pH: 5.16
Pace Analytical Services - Indianapolis

Arsenic	<0.10	mg/L	0.10	1	04/10/20 06:12	04/11/20 01:14	7440-38-2	
Barium	<5.0	mg/L	5.0	1	04/10/20 06:12	04/11/20 01:14	7440-39-3	
Cadmium	<0.050	mg/L	0.050	1	04/10/20 06:12	04/11/20 01:14	7440-43-9	
Chromium	<0.10	mg/L	0.10	1	04/10/20 06:12	04/11/20 01:14	7440-47-3	
Lead	<0.10	mg/L	0.10	1	04/10/20 06:12	04/11/20 01:14	7439-92-1	
Selenium	<0.10	mg/L	0.10	1	04/10/20 06:12	04/11/20 01:14	7782-49-2	
Silver	<0.10	mg/L	0.10	1	04/10/20 06:12	04/11/20 01:14	7440-22-4	
Zinc	2.8	mg/L	0.20	1	04/10/20 06:12	04/11/20 01:14	7440-66-6	

6020 MET ICPMS

Analytical Method: EPA 6020 Preparation Method: EPA 3050B
Pace Analytical Services - Indianapolis

Arsenic	15.1	mg/kg	0.58	1	04/06/20 00:53	04/06/20 12:35	7440-38-2	
Cadmium	1.5	mg/kg	0.29	1	04/06/20 00:53	04/06/20 12:35	7440-43-9	
Chromium	104	mg/kg	1.2	1	04/06/20 00:53	04/06/20 12:35	7440-47-3	
Copper	868	mg/kg	5.8	20	04/06/20 00:53	04/06/20 10:54	7440-50-8	
Lead	20.4	mg/kg	2.9	5	04/06/20 00:53	04/06/20 10:41	7439-92-1	
Molybdenum	23.8	mg/kg	0.58	1	04/06/20 00:53	04/06/20 12:35	7439-98-7	N2
Nickel	26.6	mg/kg	0.29	1	04/06/20 00:53	04/06/20 12:35	7440-02-0	
Selenium	5.1	mg/kg	0.58	1	04/06/20 00:53	04/06/20 12:35	7782-49-2	
Silver	2.4	mg/kg	0.29	1	04/06/20 00:53	04/06/20 12:35	7440-22-4	
Zinc	1140	mg/kg	57.6	20	04/06/20 00:53	04/06/20 10:54	7440-66-6	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50253325

Sample: Digester #3 **Lab ID: 50253325001** Collected: 03/30/20 13:45 Received: 03/31/20 09:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
7470 Mercury, TCLP								
Analytical Method: EPA 7470 Preparation Method: EPA 7470								
Leachate Method/Date: EPA 1311; 04/07/20 21:10 Initial pH: 8.35; Final pH: 5.16								
Pace Analytical Services - Indianapolis								
Mercury	<0.0020	mg/L	0.0020	1	04/11/20 14:50	04/13/20 08:56	7439-97-6	
7471 Mercury								
Analytical Method: EPA 7471 Preparation Method: EPA 7471								
Pace Analytical Services - Indianapolis								
Mercury	<6.0	mg/kg	6.0	1	04/10/20 16:01	04/11/20 16:20	7439-97-6	
8270 SVOC SS Soil								
Analytical Method: EPA 8270 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
Acenaphthene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	83-32-9	
Acenaphthylene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	208-96-8	
Anthracene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	120-12-7	
Benzo(a)anthracene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	56-55-3	
Benzo(a)pyrene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	50-32-8	
Benzo(b)fluoranthene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	205-99-2	
Benzo(g,h,i)perylene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	191-24-2	
Benzo(k)fluoranthene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	207-08-9	
Benzyl alcohol	<114000	ug/kg	114000	1	04/10/20 11:05	04/10/20 17:38	100-51-6	
4-Bromophenylphenyl ether	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	101-55-3	
Butylbenzylphthalate	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	85-68-7	
4-Chloro-3-methylphenol	<114000	ug/kg	114000	1	04/10/20 11:05	04/10/20 17:38	132-64-9	
4-Chloroaniline	<114000	ug/kg	114000	1	04/10/20 11:05	04/10/20 17:38	106-47-8	
bis(2-Chloroethoxy)methane	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	111-91-1	
bis(2-Chloroethyl) ether	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	111-44-4	
bis(2chloro1methylethyl) ether	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	108-60-1	
2-Chloronaphthalene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	91-58-7	
2-Chlorophenol	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	95-57-8	
4-Chlorophenylphenyl ether	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	7005-72-3	
Chrysene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	218-01-9	
Dibenz(a,h)anthracene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	53-70-3	
Dibenzofuran	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	132-64-9	
3,3'-Dichlorobenzidine	<114000	ug/kg	114000	1	04/10/20 11:05	04/10/20 17:38	91-94-1	
2,4-Dichlorophenol	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	120-83-2	
Diethylphthalate	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	84-66-2	
2,4-Dimethylphenol	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	105-67-9	
Dimethylphthalate	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	131-11-3	
Di-n-butylphthalate	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	84-74-2	
4,6-Dinitro-2-methylphenol	<114000	ug/kg	114000	1	04/10/20 11:05	04/10/20 17:38	534-52-1	
2,4-Dinitrophenol	<276000	ug/kg	276000	1	04/10/20 11:05	04/10/20 17:38	51-28-5	
2,4-Dinitrotoluene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	121-14-2	
2,6-Dinitrotoluene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	606-20-2	
Di-n-octylphthalate	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	117-84-0	
bis(2-Ethylhexyl)phthalate	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	117-81-7	
Fluoranthene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	206-44-0	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50253325

Sample: Digester #3 **Lab ID: 50253325001** Collected: 03/30/20 13:45 Received: 03/31/20 09:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270 SVOC SS Soil								
Analytical Method: EPA 8270 Preparation Method: EPA 3546								
Pace Analytical Services - Indianapolis								
Fluorene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	86-73-7	
Hexachloro-1,3-butadiene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	87-68-3	
Hexachlorobenzene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	118-74-1	
Hexachlorocyclopentadiene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	77-47-4	
Hexachloroethane	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	67-72-1	
Indeno(1,2,3-cd)pyrene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	193-39-5	
Isophorone	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	78-59-1	
1-Methylnaphthalene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	90-12-0	
2-Methylnaphthalene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	91-57-6	
2-Methylphenol(o-Cresol)	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	95-48-7	
3&4-Methylphenol(m&p Cresol)	<114000	ug/kg	114000	1	04/10/20 11:05	04/10/20 17:38		
Naphthalene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	91-20-3	
2-Nitroaniline	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	88-74-4	
3-Nitroaniline	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	99-09-2	
4-Nitroaniline	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	100-01-6	
Nitrobenzene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	98-95-3	
2-Nitrophenol	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	88-75-5	
4-Nitrophenol	<276000	ug/kg	276000	1	04/10/20 11:05	04/10/20 17:38	100-02-7	
N-Nitroso-di-n-propylamine	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	621-64-7	
N-Nitrosodiphenylamine	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	86-30-6	
Pentachlorophenol	<276000	ug/kg	276000	1	04/10/20 11:05	04/10/20 17:38	87-86-5	
Phenanthrene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	85-01-8	
Phenol	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	108-95-2	
Pyrene	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	129-00-0	
2,4,5-Trichlorophenol	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	95-95-4	
2,4,6-Trichlorophenol	<56900	ug/kg	56900	1	04/10/20 11:05	04/10/20 17:38	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	67	%.	26-96	1	04/10/20 11:05	04/10/20 17:38	4165-60-0	
Phenol-d5 (S)	73	%.	32-93	1	04/10/20 11:05	04/10/20 17:38	4165-62-2	
2-Fluorophenol (S)	75	%.	24-95	1	04/10/20 11:05	04/10/20 17:38	367-12-4	
2,4,6-Tribromophenol (S)	74	%.	20-109	1	04/10/20 11:05	04/10/20 17:38	118-79-6	
2-Fluorobiphenyl (S)	71	%.	36-91	1	04/10/20 11:05	04/10/20 17:38	321-60-8	
p-Terphenyl-d14 (S)	113	%.	27-117	1	04/10/20 11:05	04/10/20 17:38	1718-51-0	

8270 MSSV TCLP Sep Funnel

Analytical Method: EPA 8270 Preparation Method: EPA 3510

Leachate Method/Date: EPA 1311; 04/07/20 21:10 Initial pH: 8.35; Final pH: 5.16

Pace Analytical Services - Indianapolis

1,4-Dichlorobenzene	<0.10	mg/L	0.10	1	04/14/20 20:58	04/15/20 12:17	106-46-7	
2,4-Dinitrotoluene	<0.10	mg/L	0.10	1	04/14/20 20:58	04/15/20 12:17	121-14-2	
Hexachloro-1,3-butadiene	<0.10	mg/L	0.10	1	04/14/20 20:58	04/15/20 12:17	87-68-3	
Hexachlorobenzene	<0.10	mg/L	0.10	1	04/14/20 20:58	04/15/20 12:17	118-74-1	
Hexachloroethane	<0.10	mg/L	0.10	1	04/14/20 20:58	04/15/20 12:17	67-72-1	
2-Methylphenol(o-Cresol)	<0.10	mg/L	0.10	1	04/14/20 20:58	04/15/20 12:17	95-48-7	
3&4-Methylphenol(m&p Cresol)	<0.20	mg/L	0.20	1	04/14/20 20:58	04/15/20 12:17		
Nitrobenzene	<0.10	mg/L	0.10	1	04/14/20 20:58	04/15/20 12:17	98-95-3	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50253325

Sample: Digester #3 **Lab ID: 50253325001** Collected: 03/30/20 13:45 Received: 03/31/20 09:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
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8270 MSSV TCLP Sep Funnel

Analytical Method: EPA 8270 Preparation Method: EPA 3510

Leachate Method/Date: EPA 1311; 04/07/20 21:10 Initial pH: 8.35; Final pH: 5.16

Pace Analytical Services - Indianapolis

Pentachlorophenol	<0.50	mg/L	0.50	1	04/14/20 20:58	04/15/20 12:17	87-86-5	
Pyridine	<0.10	mg/L	0.10	1	04/14/20 20:58	04/15/20 12:17	110-86-1	
2,4,5-Trichlorophenol	<0.50	mg/L	0.50	1	04/14/20 20:58	04/15/20 12:17	95-95-4	
2,4,6-Trichlorophenol	<0.10	mg/L	0.10	1	04/14/20 20:58	04/15/20 12:17	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	60	%	49-100	1	04/14/20 20:58	04/15/20 12:17	4165-60-0	
2-Fluorobiphenyl (S)	56	%	42-95	1	04/14/20 20:58	04/15/20 12:17	321-60-8	
p-Terphenyl-d14 (S)	75	%	52-124	1	04/14/20 20:58	04/15/20 12:17	1718-51-0	
Phenol-d5 (S)	27	%	10-51	1	04/14/20 20:58	04/15/20 12:17	4165-62-2	
2-Fluorophenol (S)	39	%	23-70	1	04/14/20 20:58	04/15/20 12:17	367-12-4	
2,4,6-Tribromophenol (S)	78	%	54-106	1	04/14/20 20:58	04/15/20 12:17	118-79-6	

8260 MSV 5030 Low Level

Analytical Method: EPA 8260

Pace Analytical Services - Indianapolis

Acetone	<14700	mg/kg	14700	5000		04/08/20 17:45	67-64-1	
Acrolein	<14700	mg/kg	14700	5000		04/08/20 17:45	107-02-8	
Acrylonitrile	<14700	mg/kg	14700	5000		04/08/20 17:45	107-13-1	
Benzene	<733	mg/kg	733	5000		04/08/20 17:45	71-43-2	
Bromodichloromethane	<733	mg/kg	733	5000		04/08/20 17:45	75-27-4	
Bromoform	<733	mg/kg	733	5000		04/08/20 17:45	75-25-2	
Bromomethane	<733	mg/kg	733	5000		04/08/20 17:45	74-83-9	
2-Butanone (MEK)	<3660	mg/kg	3660	5000		04/08/20 17:45	78-93-3	
Carbon disulfide	<1470	mg/kg	1470	5000		04/08/20 17:45	75-15-0	
Carbon tetrachloride	<733	mg/kg	733	5000		04/08/20 17:45	56-23-5	
Chlorobenzene	<733	mg/kg	733	5000		04/08/20 17:45	108-90-7	
Chloroethane	<733	mg/kg	733	5000		04/08/20 17:45	75-00-3	
Chloroform	<733	mg/kg	733	5000		04/08/20 17:45	67-66-3	
Chloromethane	<733	mg/kg	733	5000		04/08/20 17:45	74-87-3	
1,2-Dibromo-3-chloropropane	<1470	mg/kg	1470	5000		04/08/20 17:45	96-12-8	
Dibromochloromethane	<733	mg/kg	733	5000		04/08/20 17:45	124-48-1	
1,2-Dibromoethane (EDB)	<733	mg/kg	733	5000		04/08/20 17:45	106-93-4	
Dibromomethane	<733	mg/kg	733	5000		04/08/20 17:45	74-95-3	
1,2-Dichlorobenzene	<733	mg/kg	733	5000		04/08/20 17:45	95-50-1	
1,3-Dichlorobenzene	<733	mg/kg	733	5000		04/08/20 17:45	541-73-1	
1,4-Dichlorobenzene	<733	mg/kg	733	5000		04/08/20 17:45	106-46-7	
trans-1,4-Dichloro-2-butene	<14700	mg/kg	14700	5000		04/08/20 17:45	110-57-6	
Dichlorodifluoromethane	<733	mg/kg	733	5000		04/08/20 17:45	75-71-8	
1,1-Dichloroethane	<733	mg/kg	733	5000		04/08/20 17:45	75-34-3	
1,2-Dichloroethane	<733	mg/kg	733	5000		04/08/20 17:45	107-06-2	
1,1-Dichloroethene	<733	mg/kg	733	5000		04/08/20 17:45	75-35-4	
cis-1,2-Dichloroethene	<733	mg/kg	733	5000		04/08/20 17:45	156-59-2	
trans-1,2-Dichloroethene	<733	mg/kg	733	5000		04/08/20 17:45	156-60-5	
1,2-Dichloropropane	<733	mg/kg	733	5000		04/08/20 17:45	78-87-5	
cis-1,3-Dichloropropene	<733	mg/kg	733	5000		04/08/20 17:45	10061-01-5	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50253325

Sample: Digester #3 **Lab ID: 50253325001** Collected: 03/30/20 13:45 Received: 03/31/20 09:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8260 MSV 5030 Low Level		Analytical Method: EPA 8260 Pace Analytical Services - Indianapolis						
trans-1,3-Dichloropropene	<733	mg/kg	733	5000		04/08/20 17:45	10061-02-6	
Ethylbenzene	<733	mg/kg	733	5000		04/08/20 17:45	100-41-4	
Ethyl methacrylate	<14700	mg/kg	14700	5000		04/08/20 17:45	97-63-2	
2-Hexanone	<14700	mg/kg	14700	5000		04/08/20 17:45	591-78-6	
Iodomethane	<14700	mg/kg	14700	5000		04/08/20 17:45	74-88-4	
Methylene Chloride	<2930	mg/kg	2930	5000		04/08/20 17:45	75-09-2	
Methyl methacrylate	<1470	mg/kg	1470	5000		04/08/20 17:45	80-62-6	
4-Methyl-2-pentanone (MIBK)	<3660	mg/kg	3660	5000		04/08/20 17:45	108-10-1	
Methyl-tert-butyl ether	<733	mg/kg	733	5000		04/08/20 17:45	1634-04-4	
Styrene	<733	mg/kg	733	5000		04/08/20 17:45	100-42-5	
1,1,1,2-Tetrachloroethane	<733	mg/kg	733	5000		04/08/20 17:45	630-20-6	
1,1,2,2-Tetrachloroethane	<733	mg/kg	733	5000		04/08/20 17:45	79-34-5	
Tetrachloroethene	<733	mg/kg	733	5000		04/08/20 17:45	127-18-4	
Toluene	<733	mg/kg	733	5000		04/08/20 17:45	108-88-3	
1,1,1-Trichloroethane	<733	mg/kg	733	5000		04/08/20 17:45	71-55-6	
1,1,2-Trichloroethane	<733	mg/kg	733	5000		04/08/20 17:45	79-00-5	
Trichloroethene	<733	mg/kg	733	5000		04/08/20 17:45	79-01-6	
Trichlorofluoromethane	<733	mg/kg	733	5000		04/08/20 17:45	75-69-4	
1,2,3-Trichloropropane	<733	mg/kg	733	5000		04/08/20 17:45	96-18-4	
Vinyl acetate	<14700	mg/kg	14700	5000		04/08/20 17:45	108-05-4	
Vinyl chloride	<733	mg/kg	733	5000		04/08/20 17:45	75-01-4	
Xylene (Total)	<1470	mg/kg	1470	5000		04/08/20 17:45	1330-20-7	
Surrogates								
Dibromofluoromethane (S)	108	%.	73-133	5000		04/08/20 17:45	1868-53-7	2d
Toluene-d8 (S)	98	%.	73-130	5000		04/08/20 17:45	2037-26-5	
4-Bromofluorobenzene (S)	100	%.	55-129	5000		04/08/20 17:45	460-00-4	

8260 MSV TCLP

Analytical Method: EPA 5030/8260 Leachate Method/Date: EPA 1311; 04/07/20 21:10

Pace Analytical Services - Indianapolis

Benzene	<0.050	mg/L	0.050	1		04/09/20 21:33	71-43-2	
2-Butanone (MEK)	<1.0	mg/L	1.0	1		04/09/20 21:33	78-93-3	
Carbon tetrachloride	<0.050	mg/L	0.050	1		04/09/20 21:33	56-23-5	
Chlorobenzene	<0.050	mg/L	0.050	1		04/09/20 21:33	108-90-7	
Chloroform	<0.050	mg/L	0.050	1		04/09/20 21:33	67-66-3	
1,2-Dichloroethane	<0.050	mg/L	0.050	1		04/09/20 21:33	107-06-2	
1,1-Dichloroethene	<0.050	mg/L	0.050	1		04/09/20 21:33	75-35-4	
Tetrachloroethene	<0.050	mg/L	0.050	1		04/09/20 21:33	127-18-4	
Trichloroethene	<0.050	mg/L	0.050	1		04/09/20 21:33	79-01-6	
Vinyl chloride	<0.020	mg/L	0.020	1		04/09/20 21:33	75-01-4	
Surrogates								
4-Bromofluorobenzene (S)	96	%.	85-116	1		04/09/20 21:33	460-00-4	
Dibromofluoromethane (S)	107	%.	75-120	1		04/09/20 21:33	1868-53-7	
Toluene-d8 (S)	92	%.	83-111	1		04/09/20 21:33	2037-26-5	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50253325

Sample: Digester #3 **Lab ID: 50253325001** Collected: 03/30/20 13:45 Received: 03/31/20 09:45 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
Percent Moisture								
Analytical Method: SM 2540G Pace Analytical Services - Indianapolis								
Percent Moisture	96.6	%	0.10	1		04/04/20 11:49		
2540G Total Percent Solids								
Analytical Method: SM 2540G Pace Analytical Services - Indianapolis								
Total Solids	3.4	%	0.10	1		04/04/20 11:50		N2
2540G Total Volatile Solids								
Analytical Method: SM 2540G Pace Analytical Services - Indianapolis								
Total Volatile Solids	54.7	%	0.10	1		04/04/20 11:49		N2
9038 Sulfate Soil								
Analytical Method: EPA 9038 Preparation Method: EPA 9038 Pace Analytical Services - Indianapolis								
Sulfate	21800	mg/kg	14600	5	04/06/20 11:36	04/06/20 14:01	14808-79-8	N2
9045 pH Soil								
Analytical Method: EPA 9045 Pace Analytical Services - Indianapolis								
pH at 25 Degrees C	8.1	Std. Units	0.10	1		04/08/20 12:01		H3
351.2 Total Kjeldahl Nitrogen								
Analytical Method: EPA 351.2 Preparation Method: EPA 351.2 Pace Analytical Services - Indianapolis								
Nitrogen, Kjeldahl, Total	76900	mg/kg	7220	5	04/06/20 11:10	04/09/20 12:47	7727-37-9	N2
353.2 Nitrogen, NO2/NO3								
Analytical Method: EPA 353.2 Preparation Method: EPA 353.2 Pace Analytical Services - Indianapolis								
Nitrogen, NO2 plus NO3	<145	mg/kg	145	1	04/07/20 10:42	04/07/20 14:42		N2
Nitrogen, Nitrate	<145	mg/kg	145	1	04/07/20 10:42	04/07/20 14:42	14797-55-8	N2
4500 Ammonia Soil, Distilled								
Analytical Method: SM 4500-NH3 G Preparation Method: SM 4500-NH3 B Pace Analytical Services - Indianapolis								
Nitrogen, Ammonia	29200	mg/kg	1380	1	04/08/20 12:28	04/08/20 14:33	7664-41-7	N2
4500 Chloride in Soil								
Analytical Method: SM 4500-Cl-E Preparation Method: SM 4500-Cl-E Pace Analytical Services - Indianapolis								
Chloride	5750	mg/kg	2930	1	04/13/20 15:33	04/14/20 08:41	16887-00-6	N2

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

QC Batch: 556373	Analysis Method: EPA 7470
QC Batch Method: EPA 7470	Analysis Description: 7470 Mercury TCLP
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2565885 Matrix: Water

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/L	<0.00067	0.00067	04/13/20 08:50	

LABORATORY CONTROL SAMPLE: 2565886

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	0.005	0.0043	86	80-120	

MATRIX SPIKE SAMPLE: 2565887

Parameter	Units	50253325001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	<0.0020	0.015	0.014	90	75-125	

MATRIX SPIKE SAMPLE: 2565888

Parameter	Units	50253377001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.014	91	75-125	

MATRIX SPIKE SAMPLE: 2565889

Parameter	Units	50253551001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	<0.0020	0.015	0.0046	31	75-125	M0

MATRIX SPIKE SAMPLE: 2565890

Parameter	Units	50253573002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.013	89	75-125	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2565891 2565892												
Parameter	Units	50253627015	MS	MSD	MS	MSD	MS	MSD	% Rec	Max	Qual	
		Result	Spike Conc.	Spike Conc.								Result
Mercury	mg/L	ND	0.015	0.015	0.014	0.014	92	94	75-125	2	20	

MATRIX SPIKE SAMPLE: 2567820							
Parameter	Units	50254041001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	0.017	0.015	0.032	100	75-125	

MATRIX SPIKE SAMPLE: 2567821							
Parameter	Units	50253343001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.014	95	75-125	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50253325

QC Batch: 556134

Analysis Method: EPA 7471

QC Batch Method: EPA 7471

Analysis Description: 7471 Mercury

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2564778

Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/kg	<0.20	0.20	04/11/20 16:12	

LABORATORY CONTROL SAMPLE: 2564779

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	0.52	0.52	101	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2564780 2564781

Parameter	Units	50253531002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	ND	0.59	0.59	0.64	0.65	96	98	75-125	2	20	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50253325

QC Batch: 555146

Analysis Method: EPA 6010

QC Batch Method: EPA 3050

Analysis Description: 6010 MET

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2560143

Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Aluminum	mg/kg	<50.0	50.0	04/07/20 13:15	
Calcium	mg/kg	<50.0	50.0	04/07/20 13:15	
Magnesium	mg/kg	<50.0	50.0	04/07/20 13:15	
Phosphorus	mg/kg	ND	50.0	04/07/20 13:15	N2
Potassium	mg/kg	<50.0	50.0	04/07/20 13:15	
Sodium	mg/kg	<50.0	50.0	04/07/20 13:15	

LABORATORY CONTROL SAMPLE: 2560144

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	500	475	95	80-120	
Calcium	mg/kg	500	471	94	80-120	
Magnesium	mg/kg	500	486	97	80-120	
Phosphorus	mg/kg	50	53.0	106	80-120	N2
Potassium	mg/kg	500	497	99	80-120	
Sodium	mg/kg	500	487	97	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2560145 2560146

Parameter	Units	50253312003 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	6190000 ug/kg	517	553	7910	8080	332	341	75-125	2	20	P6
Calcium	mg/kg	439000 ug/kg	517	553	1030	1120	115	123	75-125	8	20	
Magnesium	mg/kg	619000 ug/kg	517	553	1300	1300	132	123	75-125	0	20	M0
Phosphorus	mg/kg	173000 ug/kg	51.7	55.3	218	230	87	104	75-125	5	20	N2
Potassium	mg/kg	140000 ug/kg	517	553	711	763	110	113	75-125	7	20	
Sodium	mg/kg	49700 ug/kg	517	553	538	581	94	96	75-125	8	20	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50253325

QC Batch: 556345

QC Batch Method: EPA 3010

Analysis Method: EPA 6010

Analysis Description: 6010 MET TCLP

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2565786

Matrix: Water

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/L	<0.010	0.010	04/11/20 01:12	
Barium	mg/L	<0.50	0.50	04/11/20 01:12	
Cadmium	mg/L	<0.0050	0.0050	04/11/20 01:12	
Chromium	mg/L	<0.010	0.010	04/11/20 01:12	
Lead	mg/L	<0.010	0.010	04/11/20 01:12	
Selenium	mg/L	<0.010	0.010	04/11/20 01:12	
Silver	mg/L	<0.010	0.010	04/11/20 01:12	
Zinc	mg/L	<0.020	0.020	04/11/20 01:12	

LABORATORY CONTROL SAMPLE: 2565787

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	1	0.95	95	80-120	
Barium	mg/L	1	0.95	95	80-120	
Cadmium	mg/L	1	0.96	96	80-120	
Chromium	mg/L	1	0.94	94	80-120	
Lead	mg/L	1	0.93	93	80-120	
Selenium	mg/L	1	0.96	96	80-120	
Silver	mg/L	0.5	0.47	95	80-120	
Zinc	mg/L	1	0.96	96	80-120	

MATRIX SPIKE SAMPLE: 2565788

Parameter	Units	50253325001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	<0.10	10	9.9	99	50-150	
Barium	mg/L	<5.0	10	9.9	97	50-150	
Cadmium	mg/L	<0.050	10	9.8	98	50-150	
Chromium	mg/L	<0.10	10	9.5	95	50-150	
Lead	mg/L	<0.10	10	9.2	92	50-150	
Selenium	mg/L	<0.10	10	10	100	50-150	
Silver	mg/L	<0.10	5	4.8	97	50-150	
Zinc	mg/L	2.8	10	12.7	98	50-150	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

MATRIX SPIKE SAMPLE:		2565789					
Parameter	Units	50253551001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	<0.10	10	10.3	102	50-150	
Barium	mg/L	<5.0	10	9.6	95	50-150	
Cadmium	mg/L	<0.050	10	9.9	99	50-150	
Chromium	mg/L	<0.10	10	9.2	92	50-150	
Lead	mg/L	<0.10	10	8.6	86	50-150	
Selenium	mg/L	3.8	10	14.0	102	50-150	
Silver	mg/L	<0.10	5	5.0	99	50-150	
Zinc	mg/L	<200 ug/L	10	9.9	97	50-150	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE:		2565790			2565791							
Parameter	Units	50253627009 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/L	ND	10	10	10.2	9.9	102	99	50-150	3	20	
Barium	mg/L	ND	10	10	10.1	9.8	100	96	50-150	3	20	
Cadmium	mg/L	ND	10	10	10.2	9.9	101	99	50-150	3	20	
Chromium	mg/L	ND	10	10	9.8	9.5	98	95	50-150	3	20	
Lead	mg/L	ND	10	10	9.6	9.3	95	92	50-150	3	20	
Selenium	mg/L	ND	10	10	10.4	10.0	103	100	50-150	3	20	
Silver	mg/L	ND	5	5	5.1	4.9	101	98	50-150	4	20	
Zinc	mg/L	86600 ug/L	10	10	98.7	97.3	122	108	50-150	1	20	

MATRIX SPIKE SAMPLE:		2565792					
Parameter	Units	50253953002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.7	97	50-150	
Barium	mg/L	ND	10	10.4	95	50-150	
Cadmium	mg/L	ND	10	9.7	97	50-150	
Chromium	mg/L	ND	10	9.3	93	50-150	
Lead	mg/L	ND	10	9.0	90	50-150	
Selenium	mg/L	ND	10	9.8	98	50-150	
Silver	mg/L	ND	5	4.8	97	50-150	
Zinc	mg/L	254 ug/L	10	9.5	93	50-150	

MATRIX SPIKE SAMPLE:		2565793					
Parameter	Units	50253831001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.6	96	50-150	
Barium	mg/L	ND	10	9.6	96	50-150	
Cadmium	mg/L	ND	10	9.6	96	50-150	
Chromium	mg/L	ND	10	9.4	94	50-150	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

MATRIX SPIKE SAMPLE:		2565793					
Parameter	Units	50253831001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Lead	mg/L	ND	10	9.3	93	50-150	
Selenium	mg/L	ND	10	9.6	96	50-150	
Silver	mg/L	ND	5	4.7	95	50-150	
Zinc	mg/L	ND	10	9.7	96	50-150	

MATRIX SPIKE SAMPLE:		2565794					
Parameter	Units	50253987001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.6	96	50-150	
Barium	mg/L	ND	10	9.7	97	50-150	
Cadmium	mg/L	ND	10	9.7	97	50-150	
Chromium	mg/L	ND	10	9.6	96	50-150	
Lead	mg/L	ND	10	9.4	94	50-150	
Selenium	mg/L	ND	10	9.8	98	50-150	
Silver	mg/L	ND	5	4.8	97	50-150	
Zinc	mg/L	ND	10	9.8	97	50-150	

MATRIX SPIKE SAMPLE:		2565795					
Parameter	Units	50253893001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.5	95	50-150	
Barium	mg/L	ND	10	9.7	95	50-150	
Cadmium	mg/L	ND	10	9.6	96	50-150	
Chromium	mg/L	ND	10	9.5	94	50-150	
Lead	mg/L	ND	10	9.3	92	50-150	
Selenium	mg/L	ND	10	9.7	97	50-150	
Silver	mg/L	ND	5	4.8	96	50-150	
Zinc	mg/L	9860 ug/L	10	19.1	93	50-150	

MATRIX SPIKE SAMPLE:		2565796					
Parameter	Units	50253908001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.5	95	50-150	
Barium	mg/L	ND	10	9.5	94	50-150	
Cadmium	mg/L	ND	10	9.5	95	50-150	
Chromium	mg/L	ND	10	9.3	93	50-150	
Lead	mg/L	ND	10	9.1	91	50-150	
Selenium	mg/L	ND	10	9.7	97	50-150	
Silver	mg/L	ND	5	4.6	92	50-150	
Zinc	mg/L	1160 ug/L	10	10.6	94	50-150	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

QC Batch: 555596	Analysis Method: EPA 6020
QC Batch Method: EPA 3050B	Analysis Description: 6020 MET
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2562771 Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	<0.10	0.10	04/06/20 09:22	
Cadmium	mg/kg	<0.050	0.050	04/06/20 09:22	
Chromium	mg/kg	<0.20	0.20	04/06/20 09:22	
Copper	mg/kg	<0.050	0.050	04/06/20 09:22	
Lead	mg/kg	<0.10	0.10	04/06/20 09:22	
Molybdenum	mg/kg	<0.10	0.10	04/06/20 09:22	N2
Nickel	mg/kg	<0.050	0.050	04/06/20 09:22	
Selenium	mg/kg	<0.10	0.10	04/06/20 09:22	
Silver	mg/kg	<0.050	0.050	04/06/20 09:22	
Zinc	mg/kg	<0.50	0.50	04/06/20 09:22	

LABORATORY CONTROL SAMPLE: 2562772

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	4	3.6	90	80-120	
Cadmium	mg/kg	4	3.8	96	80-120	
Chromium	mg/kg	4	4.0	100	80-120	
Copper	mg/kg	4	4.1	102	80-120	
Lead	mg/kg	4	4.2	105	80-120	
Molybdenum	mg/kg	4	4.0	99	80-120	N2
Nickel	mg/kg	4	4.0	99	80-120	
Selenium	mg/kg	4	3.9	96	80-120	
Silver	mg/kg	4	4.0	101	80-120	
Zinc	mg/kg	4	3.8	95	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2562773 2562774

Parameter	Units	50253312001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	<2080 ug/kg	4.3	4.3	4.6	5.3	75	92	75-125	14	20	
Cadmium	mg/kg	<208 ug/kg	4.3	4.3	4.0	4.0	92	93	75-125	0	20	
Chromium	mg/kg	7590 ug/kg	4.3	4.3	12.6	13.2	116	131	75-125	5	20	M0
Copper	mg/kg	1790 ug/kg	4.3	4.3	5.5	6.2	87	104	75-125	12	20	
Lead	mg/kg	<10400 ug/kg	4.3	4.3	8.4	11.6	105	181	75-125	32	20	1d,M0

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50253325

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2562773 2562774												
Parameter	Units	50253312001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Molybdenum	mg/kg	<1040 ug/kg	4.3	4.3	3.5	3.6	78	80	75-125	2	20	N2
Nickel	mg/kg	5230 ug/kg	4.3	4.3	9.1	9.4	90	97	75-125	3	20	
Selenium	mg/kg	312 ug/kg	4.3	4.3	3.8	3.6	82	79	75-125	5	20	
Silver	mg/kg	<104 ug/kg	4.3	4.3	4.1	4.1	95	96	75-125	0	20	
Zinc	mg/kg	25800 ug/kg	4.3	4.3	31.2	34.3	125	200	75-125	10	20	P6

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50253325

QC Batch: 556177

Analysis Method: EPA 8260

QC Batch Method: EPA 8260

Analysis Description: 8260 MSV 5030 Low

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2564964

Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
1,1,1-Trichloroethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
1,1,2,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
1,1,2-Trichloroethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
1,1-Dichloroethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
1,1-Dichloroethene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
1,2,3-Trichloropropane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
1,2-Dibromo-3-chloropropane	mg/kg	<0.010	0.010	04/08/20 17:10	
1,2-Dibromoethane (EDB)	mg/kg	<0.0050	0.0050	04/08/20 17:10	
1,2-Dichlorobenzene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
1,2-Dichloroethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
1,2-Dichloropropane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
1,3-Dichlorobenzene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
1,4-Dichlorobenzene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
2-Butanone (MEK)	mg/kg	<0.025	0.025	04/08/20 17:10	
2-Hexanone	mg/kg	<0.10	0.10	04/08/20 17:10	
4-Methyl-2-pentanone (MIBK)	mg/kg	<0.025	0.025	04/08/20 17:10	
Acetone	mg/kg	<0.10	0.10	04/08/20 17:10	
Acrolein	mg/kg	<0.10	0.10	04/08/20 17:10	
Acrylonitrile	mg/kg	<0.10	0.10	04/08/20 17:10	
Benzene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Bromodichloromethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Bromoform	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Bromomethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Carbon disulfide	mg/kg	<0.010	0.010	04/08/20 17:10	
Carbon tetrachloride	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Chlorobenzene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Chloroethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Chloroform	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Chloromethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
cis-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
cis-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Dibromochloromethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Dibromomethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Dichlorodifluoromethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Ethyl methacrylate	mg/kg	<0.10	0.10	04/08/20 17:10	
Ethylbenzene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Iodomethane	mg/kg	<0.10	0.10	04/08/20 17:10	
Methyl methacrylate	mg/kg	<0.010	0.010	04/08/20 17:10	
Methyl-tert-butyl ether	mg/kg	<0.0050	0.0050	04/08/20 17:10	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

METHOD BLANK: 2564964

Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Methylene Chloride	mg/kg	<0.020	0.020	04/08/20 17:10	
Styrene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Tetrachloroethene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Toluene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
trans-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
trans-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
trans-1,4-Dichloro-2-butene	mg/kg	<0.10	0.10	04/08/20 17:10	
Trichloroethene	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Trichlorofluoromethane	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Vinyl acetate	mg/kg	<0.10	0.10	04/08/20 17:10	
Vinyl chloride	mg/kg	<0.0050	0.0050	04/08/20 17:10	
Xylene (Total)	mg/kg	<0.010	0.010	04/08/20 17:10	
4-Bromofluorobenzene (S)	%	101	55-129	04/08/20 17:10	
Dibromofluoromethane (S)	%	107	73-133	04/08/20 17:10	
Toluene-d8 (S)	%	98	73-130	04/08/20 17:10	

LABORATORY CONTROL SAMPLE: 2564965

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1,1-Trichloroethane	mg/kg	0.05	0.048	95	72-132	
1,1,2,2-Tetrachloroethane	mg/kg	0.05	0.042	84	63-127	
1,1-Dichloroethene	mg/kg	0.05	0.049	97	70-126	
1,2-Dibromoethane (EDB)	mg/kg	0.05	0.049	98	73-124	
1,2-Dichloroethane	mg/kg	0.05	0.048	95	67-129	
1,2-Dichloropropane	mg/kg	0.05	0.049	97	72-125	
Benzene	mg/kg	0.05	0.049	97	74-117	
Chlorobenzene	mg/kg	0.05	0.048	97	72-114	
Chloroform	mg/kg	0.05	0.044	89	69-120	
cis-1,2-Dichloroethene	mg/kg	0.05	0.048	96	73-117	
Ethylbenzene	mg/kg	0.05	0.048	96	68-118	
Methyl-tert-butyl ether	mg/kg	0.05	0.047	95	75-136	
Tetrachloroethene	mg/kg	0.05	0.050	99	59-119	
Toluene	mg/kg	0.05	0.047	94	68-115	
trans-1,2-Dichloroethene	mg/kg	0.05	0.050	101	71-125	
Trichloroethene	mg/kg	0.05	0.049	97	68-118	
Vinyl chloride	mg/kg	0.05	0.049	99	49-125	
Xylene (Total)	mg/kg	0.15	0.14	96	69-118	
4-Bromofluorobenzene (S)	%			99	55-129	
Dibromofluoromethane (S)	%			100	73-133	
Toluene-d8 (S)	%			100	73-130	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

QC Batch: 556443	Analysis Method: EPA 5030/8260
QC Batch Method: EPA 5030/8260	Analysis Description: 8260 MSV TCLP
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2566359 Matrix: Water

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1-Dichloroethene	mg/L	<0.050	0.050	04/09/20 15:55	
1,2-Dichloroethane	mg/L	<0.050	0.050	04/09/20 15:55	
2-Butanone (MEK)	mg/L	<1.0	1.0	04/09/20 15:55	
Benzene	mg/L	<0.050	0.050	04/09/20 15:55	
Carbon tetrachloride	mg/L	<0.050	0.050	04/09/20 15:55	
Chlorobenzene	mg/L	<0.050	0.050	04/09/20 15:55	
Chloroform	mg/L	<0.050	0.050	04/09/20 15:55	
Tetrachloroethene	mg/L	<0.050	0.050	04/09/20 15:55	
Trichloroethene	mg/L	<0.050	0.050	04/09/20 15:55	
Vinyl chloride	mg/L	<0.020	0.020	04/09/20 15:55	
4-Bromofluorobenzene (S)	%	96	85-116	04/09/20 15:55	
Dibromofluoromethane (S)	%	113	75-120	04/09/20 15:55	
Toluene-d8 (S)	%	90	83-111	04/09/20 15:55	

LABORATORY CONTROL SAMPLE: 2566360

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	0.5	0.63	125	79-128	
1,2-Dichloroethane	mg/L	0.5	0.50	101	66-127	
2-Butanone (MEK)	mg/L	2.5	2.6	104	61-138	
Benzene	mg/L	0.5	0.49	98	75-118	
Carbon tetrachloride	mg/L	0.5	0.51	103	73-125	
Chlorobenzene	mg/L	0.5	0.50	100	80-115	
Chloroform	mg/L	0.5	0.50	100	75-117	
Tetrachloroethene	mg/L	0.5	0.50	100	70-123	
Trichloroethene	mg/L	0.5	0.50	100	78-120	
Vinyl chloride	mg/L	0.5	0.56	111	55-122	
4-Bromofluorobenzene (S)	%			98	85-116	
Dibromofluoromethane (S)	%			102	75-120	
Toluene-d8 (S)	%			97	83-111	

MATRIX SPIKE SAMPLE: 2566361

Parameter	Units	50253214001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	ND	0.5	0.63	126	60-140	
1,2-Dichloroethane	mg/L	ND	0.5	0.51	102	46-145	
2-Butanone (MEK)	mg/L	ND	2.5	2.6	105	42-150	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

MATRIX SPIKE SAMPLE:		2566361					
Parameter	Units	50253214001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Benzene	mg/L	ND	0.5	0.49	97	49-135	
Carbon tetrachloride	mg/L	ND	0.5	0.51	102	55-134	
Chlorobenzene	mg/L	ND	0.5	0.49	99	42-135	
Chloroform	mg/L	ND	0.5	0.49	99	57-130	
Tetrachloroethene	mg/L	ND	0.5	0.49	98	32-140	
Trichloroethene	mg/L	ND	0.5	0.50	99	47-137	
Vinyl chloride	mg/L	ND	0.5	0.56	112	36-136	
4-Bromofluorobenzene (S)	%				97	85-116	
Dibromofluoromethane (S)	%				102	75-120	
Toluene-d8 (S)	%				96	83-111	

MATRIX SPIKE SAMPLE:		2566362					
Parameter	Units	50253581001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	ND	0.5	0.67	134	60-140	
1,2-Dichloroethane	mg/L	ND	0.5	0.54	107	46-145	
2-Butanone (MEK)	mg/L	ND	2.5	2.8	112	42-150	
Benzene	mg/L	ND	0.5	0.52	104	49-135	
Carbon tetrachloride	mg/L	ND	0.5	0.54	107	55-134	
Chlorobenzene	mg/L	ND	0.5	0.52	103	42-135	
Chloroform	mg/L	ND	0.5	0.53	105	57-130	
Tetrachloroethene	mg/L	ND	0.5	0.52	103	32-140	
Trichloroethene	mg/L	ND	0.5	0.53	106	47-137	
Vinyl chloride	mg/L	ND	0.5	0.59	117	36-136	
4-Bromofluorobenzene (S)	%				97	85-116	
Dibromofluoromethane (S)	%				103	75-120	
Toluene-d8 (S)	%				96	83-111	

MATRIX SPIKE SAMPLE:		2566363					
Parameter	Units	50253793009 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	ND	0.5	0.60	120	60-140	
1,2-Dichloroethane	mg/L	ND	0.5	0.47	95	46-145	
2-Butanone (MEK)	mg/L	ND	2.5	2.6	105	42-150	
Benzene	mg/L	ND	0.5	0.47	94	49-135	
Carbon tetrachloride	mg/L	ND	0.5	0.49	98	55-134	
Chlorobenzene	mg/L	ND	0.5	0.47	95	42-135	
Chloroform	mg/L	ND	0.5	0.48	95	57-130	
Tetrachloroethene	mg/L	ND	0.5	0.46	93	32-140	
Trichloroethene	mg/L	ND	0.5	0.50	96	47-137	
Vinyl chloride	mg/L	ND	0.5	0.53	107	36-136	
4-Bromofluorobenzene (S)	%				96	85-116	
Dibromofluoromethane (S)	%				103	75-120	
Toluene-d8 (S)	%				95	83-111	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

MATRIX SPIKE SAMPLE:		2566364	50253804001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers	
1,1-Dichloroethene	mg/L	ND	0.5	0.63	127	60-140		
1,2-Dichloroethane	mg/L	ND	0.5	0.51	103	46-145		
2-Butanone (MEK)	mg/L	ND	2.5	2.7	108	42-150		
Benzene	mg/L	ND	0.5	0.50	99	49-135		
Carbon tetrachloride	mg/L	ND	0.5	0.52	103	55-134		
Chlorobenzene	mg/L	ND	0.5	0.49	99	42-135		
Chloroform	mg/L	ND	0.5	0.50	100	57-130		
Tetrachloroethene	mg/L	ND	0.5	0.48	97	32-140		
Trichloroethene	mg/L	ND	0.5	0.50	100	47-137		
Vinyl chloride	mg/L	ND	0.5	0.56	112	36-136		
4-Bromofluorobenzene (S)	%				97	85-116		
Dibromofluoromethane (S)	%				104	75-120		
Toluene-d8 (S)	%				97	83-111		

MATRIX SPIKE SAMPLE:		2566365	50253325001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers	
1,1-Dichloroethene	mg/L	<0.050	0.5	0.64	129	60-140		
1,2-Dichloroethane	mg/L	<0.050	0.5	0.53	105	46-145		
2-Butanone (MEK)	mg/L	<1.0	2.5	2.7	110	42-150		
Benzene	mg/L	<0.050	0.5	0.51	102	49-135		
Carbon tetrachloride	mg/L	<0.050	0.5	0.53	107	55-134		
Chlorobenzene	mg/L	<0.050	0.5	0.51	103	42-135		
Chloroform	mg/L	<0.050	0.5	0.52	104	57-130		
Tetrachloroethene	mg/L	<0.050	0.5	0.50	100	32-140		
Trichloroethene	mg/L	<0.050	0.5	0.52	105	47-137		
Vinyl chloride	mg/L	<0.020	0.5	0.58	116	36-136		
4-Bromofluorobenzene (S)	%				97	85-116		
Dibromofluoromethane (S)	%				104	75-120		
Toluene-d8 (S)	%				97	83-111		

MATRIX SPIKE SAMPLE:		2566366	50253377001	Spike	MS	MS	% Rec	
Parameter	Units	Result	Conc.	Result	% Rec	Limits	Qualifiers	
1,1-Dichloroethene	mg/L	ND	0.5	0.66	131	60-140		
1,2-Dichloroethane	mg/L	ND	0.5	0.53	106	46-145		
2-Butanone (MEK)	mg/L	ND	2.5	2.6	104	42-150		
Benzene	mg/L	ND	0.5	0.51	102	49-135		
Carbon tetrachloride	mg/L	ND	0.5	0.54	107	55-134		
Chlorobenzene	mg/L	ND	0.5	0.51	102	42-135		
Chloroform	mg/L	ND	0.5	0.52	103	57-130		
Tetrachloroethene	mg/L	ND	0.5	0.50	99	32-140		
Trichloroethene	mg/L	ND	0.5	0.51	103	47-137		
Vinyl chloride	mg/L	ND	0.5	0.58	116	36-136		

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

MATRIX SPIKE SAMPLE:		2566366					
Parameter	Units	50253377001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
4-Bromofluorobenzene (S)	%.				95	85-116	
Dibromofluoromethane (S)	%.				103	75-120	
Toluene-d8 (S)	%.				95	83-111	

MATRIX SPIKE SAMPLE:		2566367					
Parameter	Units	50253831001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	ND	0.5	0.65	130	60-140	
1,2-Dichloroethane	mg/L	ND	0.5	0.51	102	46-145	
2-Butanone (MEK)	mg/L	ND	2.5	2.6	104	42-150	
Benzene	mg/L	ND	0.5	0.50	100	49-135	
Carbon tetrachloride	mg/L	ND	0.5	0.52	105	55-134	
Chlorobenzene	mg/L	ND	0.5	0.49	98	42-135	
Chloroform	mg/L	ND	0.5	0.51	102	57-130	
Tetrachloroethene	mg/L	ND	0.5	0.47	95	32-140	
Trichloroethene	mg/L	ND	0.5	0.50	100	47-137	
Vinyl chloride	mg/L	ND	0.5	0.58	116	36-136	
4-Bromofluorobenzene (S)	%.				94	85-116	
Dibromofluoromethane (S)	%.				102	75-120	
Toluene-d8 (S)	%.				94	83-111	

MATRIX SPIKE SAMPLE:		2566442					
Parameter	Units	50253125001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	ND	0.5	0.64	129	60-140	
1,2-Dichloroethane	mg/L	ND	0.5	0.52	103	46-145	
2-Butanone (MEK)	mg/L	ND	2.5	2.4	97	42-150	
Benzene	mg/L	ND	0.5	0.51	102	49-135	
Carbon tetrachloride	mg/L	ND	0.5	0.53	106	55-134	
Chlorobenzene	mg/L	ND	0.5	0.49	99	42-135	
Chloroform	mg/L	ND	0.5	0.52	104	57-130	
Tetrachloroethene	mg/L	ND	0.5	0.48	95	32-140	
Trichloroethene	mg/L	ND	0.5	0.50	100	47-137	
Vinyl chloride	mg/L	ND	0.5	0.58	116	36-136	
4-Bromofluorobenzene (S)	%.				95	85-116	
Dibromofluoromethane (S)	%.				102	75-120	
Toluene-d8 (S)	%.				96	83-111	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50253325

QC Batch: 556391

Analysis Method: EPA 8081

QC Batch Method: EPA 3546

Analysis Description: 8081 GCS Pesticides

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2566003

Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
4,4'-DDD	ug/kg	<5.0	5.0	04/09/20 17:17	
4,4'-DDE	ug/kg	<5.0	5.0	04/09/20 17:17	
4,4'-DDT	ug/kg	<5.0	5.0	04/09/20 17:17	
Aldrin	ug/kg	<2.5	2.5	04/09/20 17:17	
alpha-BHC	ug/kg	<2.5	2.5	04/09/20 17:17	
alpha-Chlordane	ug/kg	<2.5	2.5	04/09/20 17:17	
beta-BHC	ug/kg	<2.5	2.5	04/09/20 17:17	
Chlordane (Technical)	ug/kg	<49.8	49.8	04/09/20 17:17	
delta-BHC	ug/kg	<2.5	2.5	04/09/20 17:17	
Dieldrin	ug/kg	<5.0	5.0	04/09/20 17:17	
Endosulfan I	ug/kg	<2.5	2.5	04/09/20 17:17	
Endosulfan II	ug/kg	<5.0	5.0	04/09/20 17:17	
Endosulfan sulfate	ug/kg	<5.0	5.0	04/09/20 17:17	
Endrin	ug/kg	<5.0	5.0	04/09/20 17:17	
Endrin aldehyde	ug/kg	<5.0	5.0	04/09/20 17:17	
gamma-BHC (Lindane)	ug/kg	<2.5	2.5	04/09/20 17:17	
gamma-Chlordane	ug/kg	<2.5	2.5	04/09/20 17:17	
Heptachlor	ug/kg	<2.5	2.5	04/09/20 17:17	
Heptachlor epoxide	ug/kg	<2.5	2.5	04/09/20 17:17	
Toxaphene	ug/kg	<49.8	49.8	04/09/20 17:17	
Decachlorobiphenyl (S)	%	79	10-129	04/09/20 17:17	

LABORATORY CONTROL SAMPLE: 2566004

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
4,4'-DDD	ug/kg	19.9	20.5	103	53-155	
4,4'-DDE	ug/kg	19.9	20.8	104	52-153	
4,4'-DDT	ug/kg	19.9	24.8	125	38-166	
Aldrin	ug/kg	10	10	100	49-148	
alpha-BHC	ug/kg	10	10.5	105	48-150	
alpha-Chlordane	ug/kg	10	10.4	105	52-147	
beta-BHC	ug/kg	10	9.5	96	50-150	
delta-BHC	ug/kg	10	8.0	81	35-127	
Dieldrin	ug/kg	19.9	20.5	103	52-144	
Endosulfan I	ug/kg	10	11.0	111	46-145	
Endosulfan II	ug/kg	19.9	19.6	99	52-146	
Endosulfan sulfate	ug/kg	19.9	18.9	95	50-137	
Endrin	ug/kg	19.9	20.0	101	48-149	
Endrin aldehyde	ug/kg	19.9	22.5	113	52-145	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

LABORATORY CONTROL SAMPLE: 2566004

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
gamma-BHC (Lindane)	ug/kg	10	10.3	104	47-154	
gamma-Chlordane	ug/kg	10	10.1	102	52-149	
Heptachlor	ug/kg	10	10.5	106	46-147	
Heptachlor epoxide	ug/kg	10	11.2	112	51-143	
Decachlorobiphenyl (S)	%.			91	10-129	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2566005 2566006

Parameter	Units	50253573002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
4,4'-DDD	ug/kg	ND	23.3	23.7	22.2	22.1	95	93	18-157	1	20	
4,4'-DDE	ug/kg	ND	23.3	23.7	22.8	23.3	98	98	22-154	2	20	
4,4'-DDT	ug/kg	ND	23.3	23.7	27.2	27.0	117	114	10-183	1	20	
Aldrin	ug/kg	ND	11.7	11.9	11.2	11.3	96	95	21-154	1	20	
alpha-BHC	ug/kg	ND	11.7	11.9	11.9	12.0	102	101	22-156	1	20	
alpha-Chlordane	ug/kg	ND	11.7	11.9	10.9	11.0	94	93	12-163	0	20	
beta-BHC	ug/kg	ND	11.7	11.9	11.1	11.2	95	95	18-160	1	20	
delta-BHC	ug/kg	ND	11.7	11.9	9.1	9.3	78	78	10-145	2	20	
Dieldrin	ug/kg	ND	23.3	23.7	22.9	23.1	98	97	21-147	1	20	
Endosulfan I	ug/kg	ND	11.7	11.9	10.7	10.8	92	91	10-159	1	20	
Endosulfan II	ug/kg	ND	23.3	23.7	21.6	21.8	93	92	10-152	1	20	
Endosulfan sulfate	ug/kg	ND	23.3	23.7	21.0	21.5	90	91	10-145	3	20	
Endrin	ug/kg	ND	23.3	23.7	22.3	22.4	95	95	17-149	1	20	
Endrin aldehyde	ug/kg	ND	23.3	23.7	20.9	20.9	90	88	10-154	0	20	
gamma-BHC (Lindane)	ug/kg	ND	11.7	11.9	11.9	12.1	102	102	17-165	2	20	
gamma-Chlordane	ug/kg	ND	11.7	11.9	11.2	11.2	96	94	13-156	0	20	
Heptachlor	ug/kg	ND	11.7	11.9	11.8	11.8	101	99	12-164	0	20	
Heptachlor epoxide	ug/kg	ND	11.7	11.9	10.8	10.8	92	91	18-153	0	20	
Decachlorobiphenyl (S)	%.						71	69	10-129			

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

QC Batch: 556776	Analysis Method: EPA 8081
QC Batch Method: EPA 3510	Analysis Description: 8081 GCS TCLP Pesticides
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2568116 Matrix: Water

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chlordane (Technical)	mg/L	<0.00050	0.00050	04/13/20 14:59	
Endrin	mg/L	<0.00010	0.00010	04/13/20 14:59	
gamma-BHC (Lindane)	mg/L	<0.000050	0.000050	04/13/20 14:59	
Heptachlor	mg/L	<0.000050	0.000050	04/13/20 14:59	
Heptachlor epoxide	mg/L	<0.000050	0.000050	04/13/20 14:59	
Methoxychlor	mg/L	<0.00050	0.00050	04/13/20 14:59	
Toxaphene	mg/L	<0.0010	0.0010	04/13/20 14:59	
Decachlorobiphenyl (S)	%	37	10-113	04/13/20 14:59	

LABORATORY CONTROL SAMPLE: 2568115

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Endrin	mg/L	0.002	0.0020	102	40-160	
gamma-BHC (Lindane)	mg/L	0.001	0.0010	102	34-160	
Heptachlor	mg/L	0.001	0.00051	51	12-149	
Heptachlor epoxide	mg/L	0.001	0.00093	93	38-157	
Methoxychlor	mg/L	0.01	0.012	120	38-163	
Decachlorobiphenyl (S)	%			39	10-113	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50253325

QC Batch: 556573

Analysis Method: EPA 8082

QC Batch Method: EPA 3546

Analysis Description: 8082 GCS PCB

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2567053

Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	<100	100	04/13/20 16:07	
PCB-1221 (Aroclor 1221)	ug/kg	<100	100	04/13/20 16:07	
PCB-1232 (Aroclor 1232)	ug/kg	<100	100	04/13/20 16:07	
PCB-1242 (Aroclor 1242)	ug/kg	<100	100	04/13/20 16:07	
PCB-1248 (Aroclor 1248)	ug/kg	<100	100	04/13/20 16:07	
PCB-1254 (Aroclor 1254)	ug/kg	<100	100	04/13/20 16:07	
PCB-1260 (Aroclor 1260)	ug/kg	<100	100	04/13/20 16:07	
Tetrachloro-m-xylene (S)	%	97	25-123	04/13/20 16:07	

LABORATORY CONTROL SAMPLE: 2567054

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	166	164	99	54-128	
PCB-1260 (Aroclor 1260)	ug/kg	166	162	98	36-118	
Tetrachloro-m-xylene (S)	%			115	25-123	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2567055 2567056

Parameter	Units	50253646011 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
PCB-1016 (Aroclor 1016)	ug/kg	ND	187	185	151	130	81	70	10-145	15	20	
PCB-1260 (Aroclor 1260)	ug/kg	ND	187	185	152	130	81	70	11-124	16	20	
Tetrachloro-m-xylene (S)	%						78	77	25-123			

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

QC Batch: 556766	Analysis Method: EPA 8151A
QC Batch Method: EPA 8151	Analysis Description: 8151 GCS TCLP Herbicides
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2568086 Matrix: Water

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0010	0.0010	04/14/20 11:20	
2,4-D	mg/L	<0.0010	0.0010	04/14/20 11:20	
2,4-DCAA (S)	%.	88	30-140	04/14/20 10:49	

LABORATORY CONTROL SAMPLE: 2568087

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	0.005	0.0050	100	41-141	
2,4-D	mg/L	0.005	0.0041	82	41-125	
2,4-DCAA (S)	%.			94	30-140	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50253325

QC Batch: 556504

Analysis Method: EPA 8270

QC Batch Method: EPA 3546

Analysis Description: 8270 Solid MSSV Microwave Short Spike

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2566748

Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1-Methylnaphthalene	ug/kg	<327	327	04/10/20 17:05	
2,4,5-Trichlorophenol	ug/kg	<327	327	04/10/20 17:05	
2,4,6-Trichlorophenol	ug/kg	<327	327	04/10/20 17:05	
2,4-Dichlorophenol	ug/kg	<327	327	04/10/20 17:05	
2,4-Dimethylphenol	ug/kg	<327	327	04/10/20 17:05	
2,4-Dinitrophenol	ug/kg	<1580	1580	04/10/20 17:05	
2,4-Dinitrotoluene	ug/kg	<327	327	04/10/20 17:05	
2,6-Dinitrotoluene	ug/kg	<327	327	04/10/20 17:05	
2-Chloronaphthalene	ug/kg	<327	327	04/10/20 17:05	
2-Chlorophenol	ug/kg	<327	327	04/10/20 17:05	
2-Methylnaphthalene	ug/kg	<327	327	04/10/20 17:05	
2-Methylphenol(o-Cresol)	ug/kg	<327	327	04/10/20 17:05	
2-Nitroaniline	ug/kg	<327	327	04/10/20 17:05	
2-Nitrophenol	ug/kg	<327	327	04/10/20 17:05	
3&4-Methylphenol(m&p Cresol)	ug/kg	<653	653	04/10/20 17:05	
3,3'-Dichlorobenzidine	ug/kg	<653	653	04/10/20 17:05	
3-Nitroaniline	ug/kg	<327	327	04/10/20 17:05	
4,6-Dinitro-2-methylphenol	ug/kg	<653	653	04/10/20 17:05	
4-Bromophenylphenyl ether	ug/kg	<327	327	04/10/20 17:05	
4-Chloro-3-methylphenol	ug/kg	<653	653	04/10/20 17:05	
4-Chloroaniline	ug/kg	<653	653	04/10/20 17:05	
4-Chlorophenylphenyl ether	ug/kg	<327	327	04/10/20 17:05	
4-Nitroaniline	ug/kg	<327	327	04/10/20 17:05	
4-Nitrophenol	ug/kg	<1580	1580	04/10/20 17:05	
Acenaphthene	ug/kg	<327	327	04/10/20 17:05	
Acenaphthylene	ug/kg	<327	327	04/10/20 17:05	
Anthracene	ug/kg	<327	327	04/10/20 17:05	
Benzo(a)anthracene	ug/kg	<327	327	04/10/20 17:05	
Benzo(a)pyrene	ug/kg	<327	327	04/10/20 17:05	
Benzo(b)fluoranthene	ug/kg	<327	327	04/10/20 17:05	
Benzo(g,h,i)perylene	ug/kg	<327	327	04/10/20 17:05	
Benzo(k)fluoranthene	ug/kg	<327	327	04/10/20 17:05	
Benzyl alcohol	ug/kg	<653	653	04/10/20 17:05	
bis(2-Chloroethoxy)methane	ug/kg	<327	327	04/10/20 17:05	
bis(2-Chloroethyl) ether	ug/kg	<327	327	04/10/20 17:05	
bis(2-Ethylhexyl)phthalate	ug/kg	<327	327	04/10/20 17:05	
bis(2chloro1methylethyl) ether	ug/kg	<327	327	04/10/20 17:05	
Butylbenzylphthalate	ug/kg	<327	327	04/10/20 17:05	
Chrysene	ug/kg	<327	327	04/10/20 17:05	
Di-n-butylphthalate	ug/kg	<327	327	04/10/20 17:05	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

METHOD BLANK: 2566748

Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Di-n-octylphthalate	ug/kg	<327	327	04/10/20 17:05	
Dibenz(a,h)anthracene	ug/kg	<327	327	04/10/20 17:05	
Dibenzofuran	ug/kg	<327	327	04/10/20 17:05	
Diethylphthalate	ug/kg	<327	327	04/10/20 17:05	
Dimethylphthalate	ug/kg	<327	327	04/10/20 17:05	
Fluoranthene	ug/kg	<327	327	04/10/20 17:05	
Fluorene	ug/kg	<327	327	04/10/20 17:05	
Hexachloro-1,3-butadiene	ug/kg	<327	327	04/10/20 17:05	
Hexachlorobenzene	ug/kg	<327	327	04/10/20 17:05	
Hexachlorocyclopentadiene	ug/kg	<327	327	04/10/20 17:05	
Hexachloroethane	ug/kg	<327	327	04/10/20 17:05	
Indeno(1,2,3-cd)pyrene	ug/kg	<327	327	04/10/20 17:05	
Isophorone	ug/kg	<327	327	04/10/20 17:05	
N-Nitroso-di-n-propylamine	ug/kg	<327	327	04/10/20 17:05	
N-Nitrosodiphenylamine	ug/kg	<327	327	04/10/20 17:05	
Naphthalene	ug/kg	<327	327	04/10/20 17:05	
Nitrobenzene	ug/kg	<327	327	04/10/20 17:05	
Pentachlorophenol	ug/kg	<1580	1580	04/10/20 17:05	
Phenanthrene	ug/kg	<327	327	04/10/20 17:05	
Phenol	ug/kg	<327	327	04/10/20 17:05	
Pyrene	ug/kg	<327	327	04/10/20 17:05	
2,4,6-Tribromophenol (S)	%	86	20-109	04/10/20 17:05	
2-Fluorobiphenyl (S)	%	83	36-91	04/10/20 17:05	
2-Fluorophenol (S)	%	86	24-95	04/10/20 17:05	
Nitrobenzene-d5 (S)	%	79	26-96	04/10/20 17:05	
p-Terphenyl-d14 (S)	%	133	27-117	04/10/20 17:05	S3
Phenol-d5 (S)	%	88	32-93	04/10/20 17:05	

LABORATORY CONTROL SAMPLE: 2566749

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4-Dinitrotoluene	ug/kg	1670	1680	101	41-116	
2-Chlorophenol	ug/kg	1670	1460	88	48-97	
2-Methylnaphthalene	ug/kg	1670	1430	86	51-98	
4-Chloro-3-methylphenol	ug/kg	1670	1630	98	51-108	
4-Nitrophenol	ug/kg	1670	1610	97	42-118	
Acenaphthene	ug/kg	1670	1470	88	58-94	
Acenaphthylene	ug/kg	1670	1440	86	56-94	
Anthracene	ug/kg	1670	1500	90	59-98	
Benzo(a)anthracene	ug/kg	1670	1530	92	60-103	
Benzo(a)pyrene	ug/kg	1670	1490	89	57-96	
Benzo(b)fluoranthene	ug/kg	1670	1480	89	57-104	
Benzo(g,h,i)perylene	ug/kg	1670	1430	86	56-103	
Benzo(k)fluoranthene	ug/kg	1670	1700	102	55-104	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

LABORATORY CONTROL SAMPLE: 2566749

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chrysene	ug/kg	1670	1490	89	59-105	
Dibenz(a,h)anthracene	ug/kg	1670	1490	89	54-103	
Fluoranthene	ug/kg	1670	1660	100	57-107	
Fluorene	ug/kg	1670	1560	94	58-101	
Indeno(1,2,3-cd)pyrene	ug/kg	1670	1460	87	54-104	
N-Nitroso-di-n-propylamine	ug/kg	1670	1370	82	43-94	
Naphthalene	ug/kg	1670	1380	83	51-90	
Pentachlorophenol	ug/kg	1670	<1600	89	24-118	
Phenanthrene	ug/kg	1670	1530	92	60-96	
Phenol	ug/kg	1670	1340	80	44-101	
Pyrene	ug/kg	1670	1580	95	59-108	
2,4,6-Tribromophenol (S)	%			80	20-109	
2-Fluorobiphenyl (S)	%			79	36-91	
2-Fluorophenol (S)	%			81	24-95	
Nitrobenzene-d5 (S)	%			77	26-96	
p-Terphenyl-d14 (S)	%			119	27-117	S0
Phenol-d5 (S)	%			82	32-93	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2566750 2566751

Parameter	Units	50254190009 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
2,4-Dinitrotoluene	ug/kg	ND	1770	1760	1390	1250	78	71	10-127	10	20	
2-Chlorophenol	ug/kg	ND	1770	1760	773	530	44	30	17-107	37	20	R1
2-Methylnaphthalene	ug/kg	ND	1770	1760	1290	1160	73	66	20-120	11	20	
4-Chloro-3-methylphenol	ug/kg	ND	1770	1760	1570	1390	89	79	22-117	12	20	
4-Nitrophenol	ug/kg	ND	1770	1760	<1700	<1690	6	5	10-131		20	M1
Acenaphthene	ug/kg	ND	1770	1760	1420	1270	80	72	19-115	11	20	
Acenaphthylene	ug/kg	ND	1770	1760	1390	1240	79	70	11-122	12	20	
Anthracene	ug/kg	ND	1770	1760	1430	1350	81	76	10-133	6	20	
Benzo(a)anthracene	ug/kg	ND	1770	1760	1460	1380	82	78	10-137	5	20	
Benzo(a)pyrene	ug/kg	ND	1770	1760	1460	1400	83	80	10-132	4	20	
Benzo(b)fluoranthene	ug/kg	ND	1770	1760	1450	1330	82	76	10-134	9	20	
Benzo(g,h,i)perylene	ug/kg	ND	1770	1760	1610	1510	91	86	10-138	7	20	
Benzo(k)fluoranthene	ug/kg	ND	1770	1760	1670	1520	94	86	10-140	9	20	
Chrysene	ug/kg	ND	1770	1760	1550	1460	87	83	10-150	6	20	
Dibenz(a,h)anthracene	ug/kg	ND	1770	1760	1640	1530	93	87	10-126	7	20	
Fluoranthene	ug/kg	ND	1770	1760	1490	1400	84	79	10-128	6	20	
Fluorene	ug/kg	ND	1770	1760	1520	1360	86	77	12-127	11	20	
Indeno(1,2,3-cd)pyrene	ug/kg	ND	1770	1760	1650	1530	93	87	10-125	8	20	
N-Nitroso-di-n-propylamine	ug/kg	ND	1770	1760	1230	1150	69	65	23-101	6	20	
Naphthalene	ug/kg	ND	1770	1760	1270	1140	72	65	12-120	11	20	
Pentachlorophenol	ug/kg	ND	1770	1760	<1700	<1690	13	8	10-112		20	M1
Phenanthrene	ug/kg	ND	1770	1760	1500	1400	85	79	10-125	7	20	
Phenol	ug/kg	ND	1770	1760	1240	1080	70	61	22-105	13	20	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2566750 2566751												
Parameter	Units	50254190009 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Pyrene	ug/kg	ND	1770	1760	1670	1540	94	87	10-146	8	20	
2,4,6-Tribromophenol (S)	%						10	4	20-109			S0
2-Fluorobiphenyl (S)	%						69	60	36-91			
2-Fluorophenol (S)	%						33	25	24-95			
Nitrobenzene-d5 (S)	%						66	61	26-96			
p-Terphenyl-d14 (S)	%						94	92	27-117			
Phenol-d5 (S)	%						66	57	32-93			

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

QC Batch: 557104	Analysis Method: EPA 8270
QC Batch Method: EPA 3510	Analysis Description: 8270 TCLP MSSV
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2569435 Matrix: Water

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,4-Dichlorobenzene	mg/L	<0.010	0.010	04/15/20 11:45	
2,4,5-Trichlorophenol	mg/L	<0.050	0.050	04/15/20 11:45	
2,4,6-Trichlorophenol	mg/L	<0.010	0.010	04/15/20 11:45	
2,4-Dinitrotoluene	mg/L	<0.010	0.010	04/15/20 11:45	
2-Methylphenol(o-Cresol)	mg/L	<0.010	0.010	04/15/20 11:45	
3&4-Methylphenol(m&p Cresol)	mg/L	<0.020	0.020	04/15/20 11:45	
Hexachloro-1,3-butadiene	mg/L	<0.010	0.010	04/15/20 11:45	
Hexachlorobenzene	mg/L	<0.010	0.010	04/15/20 11:45	
Hexachloroethane	mg/L	<0.010	0.010	04/15/20 11:45	
Nitrobenzene	mg/L	<0.010	0.010	04/15/20 11:45	
Pentachlorophenol	mg/L	<0.050	0.050	04/15/20 11:45	
Pyridine	mg/L	<0.010	0.010	04/15/20 11:45	
2,4,6-Tribromophenol (S)	%	67	54-106	04/15/20 11:45	
2-Fluorobiphenyl (S)	%	49	42-95	04/15/20 11:45	
2-Fluorophenol (S)	%	36	23-70	04/15/20 11:45	
Nitrobenzene-d5 (S)	%	58	49-100	04/15/20 11:45	
p-Terphenyl-d14 (S)	%	103	52-124	04/15/20 11:45	
Phenol-d5 (S)	%	22	10-51	04/15/20 11:45	

LABORATORY CONTROL SAMPLE: 2569436

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	0.05	0.023	46	18-89	
2,4,5-Trichlorophenol	mg/L	0.05	<0.050	83	57-113	
2,4,6-Trichlorophenol	mg/L	0.05	0.037	75	63-105	
2,4-Dinitrotoluene	mg/L	0.05	0.039	77	48-112	
2-Methylphenol(o-Cresol)	mg/L	0.05	0.030	60	46-93	
3&4-Methylphenol(m&p Cresol)	mg/L	0.1	0.056	56	40-88	
Hexachloro-1,3-butadiene	mg/L	0.05	0.018	36	10-94	
Hexachlorobenzene	mg/L	0.05	0.028	56	46-75	
Hexachloroethane	mg/L	0.05	0.020	39	10-87	
Nitrobenzene	mg/L	0.05	0.035	69	51-108	
Pentachlorophenol	mg/L	0.05	<0.050	74	21-115	
Pyridine	mg/L	0.05	0.016	31	10-78	
2,4,6-Tribromophenol (S)	%			73	54-106	
2-Fluorobiphenyl (S)	%			56	42-95	
2-Fluorophenol (S)	%			46	23-70	
Nitrobenzene-d5 (S)	%			67	49-100	
p-Terphenyl-d14 (S)	%			98	52-124	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50253325

LABORATORY CONTROL SAMPLE: 2569436

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Phenol-d5 (S)	%.			30	10-51	

MATRIX SPIKE SAMPLE: 2569437

Parameter	Units	50254190016 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	ND	0.5	0.24	48	19-91	
2,4,5-Trichlorophenol	mg/L	ND	0.5	<0.50	86	40-123	
2,4,6-Trichlorophenol	mg/L	ND	0.5	0.41	81	47-113	
2,4-Dinitrotoluene	mg/L	ND	0.5	0.40	80	40-112	
2-Methylphenol(o-Cresol)	mg/L	ND	0.5	0.31	61	38-97	
3&4-Methylphenol(m&p Cresol)	mg/L	ND	1	0.57	57	36-91	
Hexachloro-1,3-butadiene	mg/L	ND	0.5	0.24	48	11-98	
Hexachlorobenzene	mg/L	ND	0.5	0.30	59	38-77	
Hexachloroethane	mg/L	ND	0.5	0.21	43	10-93	
Nitrobenzene	mg/L	ND	0.5	0.35	70	44-106	
Pentachlorophenol	mg/L	ND	0.5	<0.50	89	22-126	
Pyridine	mg/L	ND	0.5	0.24	47	10-83	
2,4,6-Tribromophenol (S)	%.				79	54-106	
2-Fluorobiphenyl (S)	%.				67	42-95	
2-Fluorophenol (S)	%.				45	23-70	
Nitrobenzene-d5 (S)	%.				68	49-100	
p-Terphenyl-d14 (S)	%.				101	52-124	
Phenol-d5 (S)	%.				28	10-51	

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50253325

QC Batch: 555552

QC Batch Method: SM 2540G

Analysis Method: SM 2540G

Analysis Description: Dry Weight/Percent Moisture

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

SAMPLE DUPLICATE: 2562576

Parameter	Units	50253325001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	96.6	96.6	0	5	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

QC Batch:	555550	Analysis Method:	SM 2540G
QC Batch Method:	SM 2540G	Analysis Description:	2540G Total Solids
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

SAMPLE DUPLICATE: 2562574

Parameter	Units	50253325001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	3.4	3.4	0	10	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

QC Batch:	555551	Analysis Method:	SM 2540G
QC Batch Method:	SM 2540G	Analysis Description:	2540G Total Volatile Solids
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

SAMPLE DUPLICATE: 2562575

Parameter	Units	50253325001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Volatile Solids	%	54.7	55.2	1	10	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

QC Batch: 555661	Analysis Method: EPA 9038
QC Batch Method: EPA 9038	Analysis Description: 9038 Sulfate Soil
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2562937 Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Sulfate	mg/kg	<99.9	99.9	04/06/20 14:08	N2

LABORATORY CONTROL SAMPLE: 2562938

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Sulfate	mg/kg	200	204	102	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2562939 2562940

Parameter	Units	50253132001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Sulfate	mg/kg	1680	4230	4230	6540	6250	115	108	90-110	4	20	M0, N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

QC Batch: 556044	Analysis Method: EPA 9045
QC Batch Method: EPA 9045	Analysis Description: 9045 pH
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

SAMPLE DUPLICATE: 2564417

Parameter	Units	50253827001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	6.8	6.8	0	2	H3

SAMPLE DUPLICATE: 2564418

Parameter	Units	50253730001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	7.0	7.0	0	2	H3

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50253325

QC Batch: 555637

Analysis Method: EPA 351.2

QC Batch Method: EPA 351.2

Analysis Description: 351.2 TKN

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2562880

Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<49.3	49.3	04/09/20 12:10	N2

LABORATORY CONTROL SAMPLE: 2562881

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	498	509	102	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2562882 2562883

Parameter	Units	50253132001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	45900	2090	2070	47500	45900	75	-1	90-110	3	20	N2,P6

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50253325

QC Batch: 555858

QC Batch Method: EPA 353.2

Analysis Method: EPA 353.2

Analysis Description: 353.2 Nitrate + Nitrite

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2563566

Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Nitrate	mg/kg	<5.0	5.0	04/07/20 14:38	N2
Nitrogen, NO2 plus NO3	mg/kg	<5.0	5.0	04/07/20 14:38	N2

LABORATORY CONTROL SAMPLE: 2563567

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Nitrate	mg/kg	10	10.8	108	90-110	N2
Nitrogen, NO2 plus NO3	mg/kg	20	21.1	105	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2563568 2563569

Parameter	Units	50253325001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Nitrate	mg/kg	<145	290	290	325	331	109	111	90-110	2	20	N2
Nitrogen, NO2 plus NO3	mg/kg	<145	580	580	643	651	107	108	90-110	1	20	N2

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids

Pace Project No.: 50253325

QC Batch: 556082

QC Batch Method: SM 4500-NH3 B

Analysis Method: SM 4500-NH3 G

Analysis Description: 4500 Ammonia, Distilled

Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2564521

Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Ammonia	mg/kg	<4.9	4.9	04/08/20 14:27	N2

LABORATORY CONTROL SAMPLE: 2564522

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Ammonia	mg/kg	20	21.2	106	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2564523 2564524

Parameter	Units	50253132001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Ammonia	mg/kg	2790	84.2	83.8	3050	3080	309	344	90-110	1	20	N2,P6

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50253325

QC Batch: 556834	Analysis Method: SM 4500-Cl-E
QC Batch Method: SM 4500-Cl-E	Analysis Description: 4500 Chloride
	Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50253325001

METHOD BLANK: 2568388 Matrix: Solid

Associated Lab Samples: 50253325001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	<99.2	99.2	04/14/20 08:39	N2

LABORATORY CONTROL SAMPLE: 2568389

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	198	193	97	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 2568390 2568391

Parameter	Units	50253784017 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	2350	1100	1100	3700	3620	98	91	90-110	2	20	N2

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QUALIFIERS

Project: Biosolids
Pace Project No.: 50253325

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
 ND - Not Detected at or above adjusted reporting limit.
 TNTC - Too Numerous To Count
 J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
 MDL - Adjusted Method Detection Limit.
 PQL - Practical Quantitation Limit.
 RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
 S - Surrogate
 1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
 Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
 LCS(D) - Laboratory Control Sample (Duplicate)
 MS(D) - Matrix Spike (Duplicate)
 DUP - Sample Duplicate
 RPD - Relative Percent Difference
 NC - Not Calculable.
 SG - Silica Gel - Clean-Up
 U - Indicates the compound was analyzed for, but not detected.
 N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
 Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
 TNI - The NELAC Institute.

ANALYTE QUALIFIERS

1d	RPD value is outside laboratory control limits due to sample non-homogeneity.
2d	Sample was analyzed at a high dilution due to matrix being oil. TMW 04-12-20
H3	Sample was received or analysis requested beyond the recognized method holding time.
M0	Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.
M1	Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery.
N2	The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
P6	Matrix spike recovery was outside laboratory control limits due to a parent sample concentration notably higher than the spike level.
R1	RPD value was outside control limits.
S0	Surrogate recovery outside laboratory control limits.
S3	Surrogate recovery exceeded laboratory control limits. Analyte presence below reporting limits in associated sample.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Biosolids

Pace Project No.: 50253325

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50253325001	Digester #3	EPA 3546	556391	EPA 8081	556453
50253325001	Digester #3	EPA 3510	556776	EPA 8081	556820
50253325001	Digester #3	EPA 3546	556573	EPA 8082	556733
50253325001	Digester #3	EPA 8151	556766	EPA 8151A	556963
50253325001	Digester #3	EPA 3050	555146	EPA 6010	555838
50253325001	Digester #3	EPA 3010	556345	EPA 6010	556653
50253325001	Digester #3	EPA 3050B	555596	EPA 6020	555608
50253325001	Digester #3	EPA 7470	556373	EPA 7470	556724
50253325001	Digester #3	EPA 7471	556134	EPA 7471	556667
50253325001	Digester #3	EPA 3546	556504	EPA 8270	556629
50253325001	Digester #3	EPA 3510	557104	EPA 8270	557219
50253325001	Digester #3	EPA 8260	556177		
50253325001	Digester #3	EPA 5030/8260	556443		
50253325001	Digester #3	SM 2540G	555552		
50253325001	Digester #3	SM 2540G	555550		
50253325001	Digester #3	SM 2540G	555551		
50253325001	Digester #3	EPA 9038	555661	EPA 9038	555677
50253325001	Digester #3	EPA 9045	556044		
50253325001	Digester #3	EPA 351.2	555637	EPA 351.2	556353
50253325001	Digester #3	EPA 353.2	555858	EPA 353.2	555901
50253325001	Digester #3	SM 4500-NH3 B	556082	SM 4500-NH3 G	556146
50253325001	Digester #3	SM 4500-CI-E	556834	SM 4500-CI-E	556913

REPORT OF LABORATORY ANALYSIS

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Sample Conditions Upon Receipt Form (SCUR)

WO# : 50253325

PM: MSB Due Date: 04/14/20
CLIENT: GR-Cadillac

Date/Time: 3-31-20 1000	Evaluated by: AWS
Client: CITY OF CADILLAC	
Project Manager: MSB	Profile ID:
Rush TAT Requested: YES ☒ NO ☐	Due Date:

Lab Notified of Rush or Short Holds: YES ☐ NO ☒	Non Conformance Form Required: YES ☐ NO ☒
Samples Received Via: FedEx ☐ UPS ☒ Client ☐ Pace Courier ☐ Other: _____	Comments:

Custody Seals Present and Intact:	YES	NO	N/A
Received Sample Information Form(s): Drinking Waters Only	YES	NO	N/A
USDA Regulated Soils: (AL, AR, CA, FL, GA, ID, LA, MS, NM, NY, NC, OK, OR, SC, TN, TX, WA or Puerto Rico)	YES	NO	N/A
Short Holds Present (< 72 Hours):	YES	NO	
Samples Received in Hold:	YES	NO	
Custody Signatures Present:	YES	NO	
Collector Signature Present:	YES	NO	
Packing Material Used:	YES	NO	
Samples Collected Today and On Ice:	YES	NO	N/A

IR Gun #: 280 281	Digital Thermometer #: 282 283
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Ice Type: WET Bagged / WET Loose BLUE NONE	1. Cooler Temp Upon Receipt: 4.2/4.7 °C
Ice Location: TOP ☒ BOTTOM MIDDLE DISPERSED	Temp should be 0-6°C (Initial/Corrected)

Temp Blank Received:	YES	NO	
Containers Intact:	YES	NO	
Correct Containers:	YES	NO	
Sufficient Volume:	YES	NO	
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation Exceptions are VOA, coliform, LLHg, O&G, or any container with a septum cap or preserved with HCl	YES	NO	N/A
Residual Chlorine Absent: (SVOC/Pest 625, PCB 608, Total/Amenable/Available Cyanide)	YES	NO	N/A
VOA Headspace Acceptable (<6mm):	YES	NO	N/A
Trip Blank Received: HCl MeOH TSP OTHER	YES	NO	

Comments: N/A	2. Cooler Temp Upon Receipt: _____ °C
	3. Cooler Temp Upon Receipt: _____ °C
	4. Cooler Temp Upon Receipt: _____ °C